HAL Helicopter Performance — Calculation Review Sheet

Hindustan Aeronautics Limited · Offline Power Calculation App

Reviewer: _____ **Designation:** _____ **Date:** _____

App Version: v1.0.0

How to use this sheet: For each value / formula, tick the checkbox if it is correct. If it is wrong, leave the box unticked and write the corrected value or formula in the *Remarks* column. Return the signed sheet to the development team.

1. AIRCRAFT DEFAULT VALUES

Source file: `src/config/logic.js → DEFAULT_AIRCRAFT`

1.1 Chetak

✓	PARAMETER	UNIT	DEFAULT	REMARKS / CORRECTED VALUE
☐	Empty Weight	lb	2 645	
☐	Maximum All-Up Weight (MAUW)	lb	4 630	
☐	Default Crew Weight	kg	180	
☐	Default Fuel	litres	575	
☐	Default Additional Load	kg	0	
☐	Default Load (Pax + cargo)	kg	75	
☐	Default Elevation	ft	0	
☐	Default QNH	mb	1 013	
☐	Default Temperature (OAT)	°C	15	

1.2 Cheetah

✓	PARAMETER	UNIT	DEFAULT	REMARKS / CORRECTED VALUE
☐	Empty Weight	lb	2 469	
☐	Maximum All-Up Weight (MAUW)	lb	4 299	
☐	Default Crew Weight	kg	180	
☐	Default Fuel	litres	500	
☐	Default Additional Load	kg	0	
☐	Default Load	kg	75	
☐	Default Elevation	ft	0	
☐	Default QNH	mb	1 013	
☐	Default Temperature (OAT)	°C	15	

1.3 Cheetal

✓	PARAMETER	UNIT	DEFAULT	REMARKS / CORRECTED VALUE
☐	Empty Weight	lb	2 756	
☐	Maximum All-Up Weight (MAUW)	lb	4 960	
☐	Default Crew Weight	kg	180	
☐	Default Fuel	litres	575	
☐	Default Additional Load	kg	0	
☐	Default Load	kg	75	
☐	Default Elevation	ft	0	
☐	Default QNH	mb	1 013	
☐	Default Temperature (OAT)	°C	15	

2. INPUT FIELD UNITS

✓	INPUT FIELD	PRIMARY UNIT	TOGGLE	REMARKS
☐	Elevation	ft	m	
☐	QNH	mb (hPa)	inHg	
☐	Temperature	°C	°F	
☐	AC Weight	lb	—	
☐	Crew Weight	kg	—	
☐	Fuel	litres	—	
☐	Additional Load	kg	—	
☐	Load (Pax + cargo)	kg	—	

3. CALCULATION FORMULAS

Source file: `src/config/logic.js → DEFAULT_FORMULAS`. Each formula is a plain-JS expression evaluated with the variables shown.

☐ **3.1 Pressure Altitude (PA) — output: ft**

```
PA = elevation + (1013 − qnh) × 30
```

variables: *elevation* (ft), *qnh* (mb)

Remarks / Corrected formula:

☐ **3.2 ISA Temperature — output: °C**

```
ISA_TEMP = 15 − (elevation / 1000) × 2
```

variables: *elevation* (ft)

Remarks / Corrected formula:

☐ **3.3 Density Altitude (DA) — output: ft**

```
DA = PA + 120 × (OAT − ISA_TEMP)
```

variables: *pa* (ft), *oat* (°C), *isa* (°C)

Remarks / Corrected formula:

☐ **3.4 Absolute Temperature — output: Kelvin**

```
ABS_TEMP = 273 + OAT
```

variables: *oat* (°C)

Remarks / Corrected formula:

☐ **3.5 All Up Weight (AUW) — output: lb**

```
AUW = ac_weight + (crew × 2.2) + (fuel × 0.775 × 2.2) + (add_load × 2.2) + (load × 2.2)
```

variables: *ac_weight* (lb), *crew* (kg), *fuel* (litres), *add_load* (kg), *load* (kg)
factors: 2.2 = kg→lb · 0.775 = aviation-fuel density (kg / litre)

Remarks / Corrected formula:

☐ **3.6 Power Available (Max Pwr) — output: fraction 0..1 (capped at 1.0)**

```
POWER_AVAIL = min(1.0, ((DA / 660) × 0.01) + 0.75)
```

variables: *density_alt* (ft)

Remarks / Corrected formula:

☐ **3.7 Power Required (Hover Pwr) — output: fraction**

```
POWER_REQ = (AUW / 6600) × (1013 / QNH) × (ABS_TEMP / 288)
```

variables: *auw* (lb), *qnh* (mb), *abs_temp* (K)

Remarks / Corrected formula:

4. DERIVED QUANTITIES & PRESENTATION

✓	QUANTITY	UNIT	FORMULA	REMARKS
☐	Power Balance %	%	`(Pwr_Avail − Pwr_Req) / Pwr_Avail × 100`	
☐	AUW Margin	lb	`MAUW − AUW`	
☐	Payload Margin	lb	`min(AUW_Margin, AUW × pwr_headroom)`	

5. FIT-TO-FLY THRESHOLDS (0.01 PRECISION)

✓	RULE	CONDITION FOR NOT FIT	REMARKS / CORRECTED RULE
☐	Weight check	`AUW − MAUW > 0.01 lb`	
☐	Power check	`POWER_REQ − POWER_AVAIL > 0.01`	
☐	Service ceiling	`DA > 18 000 ft`	

6. SAMPLE CALCULATIONS (PLEASE VERIFY)

CASE A · CHETAK, SL ISA		CASE B · CHETAK, 10 000 FT · 30 °C	
Inputs	Elev 0 ft · QNH 1013 · Temp 15 °C · AC 2 645 lb · Crew 180 kg · Fuel 575 L · Load 75 kg · Add 0	**Inputs**	Elev 10 000 ft · QNH 1013 · Temp 30 °C · AC 2 645 lb · Crew 180 kg · Fuel 575 L · Load 200 kg · Add 0
PA	0 ft	PA	10 000 ft
ISA Temp	15 °C	ISA Temp	−5 °C
DA	0 ft	DA	14 200 ft
Abs Temp	288 K	Abs Temp	303 K
AUW	4 186.38 lb	AUW	4 461.38 lb
Power Available	0.750	Power Available	0.965
Power Required	0.634	Power Required	0.711
Status	**FIT** (+15 %)	Status	**FIT** (+26 %)

Overall remarks on sample calculations:

7. OVERALL REVIEW COMMENTS

Reviewer Signature

Date

HAL Helicopter Performance App · Calculation Review Sheet · generated from `src/config/logic.js`